Spiking Free-Energy State Estimator

A complete derivation: free-energy principle, spiking inference, innovation-whitening precision learning, assumptions, and lessons learned

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Motivation and Foundational Idea

The free-energy principle, in its computational-neuroscience formulation, holds that a self-organising system minimises a quantity called *variational free energy* — an upper bound on the surprise of its sensory observations — by continuously adjusting its internal states. When the internal representation is a population of spiking neurons, this translates into a beautifully simple rule: *a neuron fires if and only if doing so reduces the free energy of the whole population*. This is the Spiking Free-Energy Constrainer (SFEC) principle.

Two largely separate problems live inside any such estimator.

1. **The fast loop.** Given that the precision (inverse variance) parameters are fixed, can the spiking population compute the correct Bayesian posterior mean?

2. **The slow loop.** Can the network *learn* those precision parameters online, from the data stream only, without ever being told the true noise statistics?

The present document derives both loops from first principles and describes the innovation-whitening approach that solves the slow loop. It also catalogues the several earlier approaches that were found to fail, together with the reasons they fail.

System Setup

2.1 The plant

Consider a continuous-time linear plant with Euler-discretised observations:

$$\dot{x} = A_{\text{int}}\, x + d_w, \qquad y_t = C\, x_t + d_\eta$$

where $x \in \mathbb{R}^n$ is the hidden state, $y_t \in \mathbb{R}^m$ is the discrete-time observation at step t, and

$$d_w \sim \mathcal{N}(0,\, Q_c\, \Delta t), \qquad d_\eta \sim \mathcal{N}(0,\, R_c\, \Delta t)$$

are i.i.d. zero-mean process and observation noise increments. The matrices Q_c and R_c are the *true* continuous-time noise covariances; the estimator does *not* have access to them.

> ***Discretisation convention.*** *The observation noise is written as $d_\eta \sim \mathcal{N}(0, R_c\, \Delta t)$, which treats $R_c \Delta t$ as the discrete-time measurement covariance. This convention corresponds to a zero-order-hold (ZOH) observation model in which the sensor integrates the signal over the interval Δt and the noise power therefore scales with Δt. An alternative convention — common when the continuous observation is modelled as a white-noise-driven ODE — places observation noise as $R_c / \Delta t$, which grows as $\Delta t \to 0$. Throughout this document we use the ZOH convention ($R_d = R_c\, \Delta t$, $Q_d = Q_c\, \Delta t$); comparisons with other derivations should check which convention they adopt.*

2.2 The neural code

The internal state estimate $\boldsymbol{\mu} \in \mathbb{R}^n$ is represented as a weighted superposition of fixed decoder columns:

$$\boldsymbol{\mu} = D\mathbf{r}, \qquad D \in \mathbb{R}^{n \times N}$$

where $\mathbf{r} \in \mathbb{R}^N$ is the vector of filtered spike counts and D is the fixed decoder matrix. The network is *overcomplete*: $N \gg n$. Each column D_i is the preferred decoding direction of neuron i.

Spike counts evolve according to leaky integration:

$$\dot{\mathbf{r}} = -\lambda\, \mathbf{r} + \mathbf{s}$$

where $\lambda > 0$ is the membrane *leak rate* (units: s^{-1}) and $\mathbf{s} \in \{0,1\}^N$ is the instantaneous spike vector. The associated membrane time constant is $\tau_r = 1/\lambda$. Throughout the fast loop we use λ consistently as a rate; the slow-loop traces use separate time-constant parameters τ_0, τ_1, τ_d, and τ_λ written in the standard form $\tau\dot{c} = -c + x$ (i.e. with dimension of time on the left-hand side). These two conventions are never mixed within the same equation.

2.3 The predictive prior

At the start of each outer time-step, the network forms a *predictive prior* by propagating the previous posterior estimate through the (approximately known) linear dynamics:

$$\boldsymbol{\mu}_t^- = \Phi\, \boldsymbol{\mu}_{t-1}, \qquad \Phi = e^{A_{\mathrm{int}}\,\Delta t} \approx I + \Delta t\, A_{\mathrm{int}}$$

This prior mean is held *frozen* during the inner spike-relaxation window; hence $\dot{\boldsymbol{\mu}}^- = 0$ within one time-step.

2.4 Matched-channel assumption

Throughout the main derivation we impose:

$$C = I, \qquad m = n$$

i.e. every state dimension is observed directly. This is called the *matched-channel* (or *full-state-observation*) case. Section 10 discusses what it would take to relax this.

THE FREE-ENERGY FUNCTIONAL

3.1 Definition

The variational free energy for the estimator is defined as a precision-weighted sum of two prediction errors plus an optional spike-cost term:

Estimator free energy

$$\mathcal{F}(\boldsymbol{\mu}, \mathbf{r}) = \underbrace{(y - C\boldsymbol{\mu})^\top \Pi_y (y - C\boldsymbol{\mu})}_{\text{sensory prediction error}} + \underbrace{(\boldsymbol{\mu}^- - \boldsymbol{\mu})^\top \Pi_\mu (\boldsymbol{\mu}^- - \boldsymbol{\mu})}_{\text{prior prediction error}} + \beta\, \mathbf{r}^\top \mathbf{r}$$

Here $\Pi_y \in \mathbb{R}^{m\times m}$ is the *sensory precision matrix* (inverse of the observation noise covariance), $\Pi_\mu \in \mathbb{R}^{n\times n}$ is the *prior precision matrix* (inverse of the one-step predictive error covariance), and $\beta \geq 0$ is a spike-cost coefficient that encourages sparse representations.

> ***Role of this functional in precision learning.*** *A fully Bayesian variational free energy over* both *states and precision matrices includes log-determinant terms of the form* $-\log|\Pi_y| - \log|\Pi_\mu|$. *These terms are necessary to prevent degenerate solutions when optimising over precisions directly. The present functional* omits *those terms deliberately:* $\mathcal{F}$ *is used only for the* fast loop, *which holds precisions fixed. The* slow loop *is* not *obtained by differentiating* $\mathcal{F}$ *with respect to* Π_y *or* Π_μ*; it is an entirely separate statistical identification procedure based on innovation whitening (Section 8). The two loops are thus decoupled by design: the fast loop minimises the quadratic free energy for frozen precisions, and the slow loop identifies the correct precisions from the temporal structure of the innovation stream.*

Define the compound precision-weighted measurement matrix and information vector:

$$M = C^\top \Pi_y C + \Pi_\mu, \qquad \mathbf{q}_t = C^\top \Pi_y y_t + \Pi_\mu\, \boldsymbol{\mu}_t^-$$

Setting $\beta = 0$ and minimising $\mathcal{F}$ over $\boldsymbol{\mu}$:

$$\frac{\partial \mathcal{F}}{\partial \boldsymbol{\mu}} = -2C^\top \Pi_y (y - C\boldsymbol{\mu}) + 2\Pi_\mu(\boldsymbol{\mu} - \boldsymbol{\mu}^-) = 0 \implies \boldsymbol{\mu}^\star = M^{-1}\mathbf{q}_t$$

This is exactly the *information-form Bayesian fusion* of measurement and prior — identical to one step of a Kalman filter.

3.2 Bayesian interpretation

With $C = I$ and diagonal precisions $\Pi_y = \pi_y I$, $\Pi_\mu = \pi_\mu I$, the posterior collapses to a precision-weighted mean in each channel independently:

$$\mu_{t,k}^\star = \frac{\pi_y\, y_{t,k} + \pi_\mu\, \mu_{t,k}^-}{\pi_y + \pi_\mu} = K y_{t,k} + (1-K)\mu_{t,k}^-, \qquad K = \frac{\pi_y}{\pi_y + \pi_\mu}$$

The scalar $K \in (0,1)$ is the *Kalman gain*: the fraction of the innovation that is absorbed into the estimate. Its theoretically optimal value is

$$K^\star = \frac{P^-}{P^- + R}, \qquad P^- = \text{prior variance}, \quad R = \text{sensor noise variance}$$

The entire precision learning problem reduces to identifying this single scalar K (per channel) without knowing P^- or R explicitly.

4.1 Single-spike free-energy change

When neuron i fires, the network state changes as $\mathbf{r} \to \mathbf{r} + e_i$ and consequently $\boldsymbol{\mu} \to \boldsymbol{\mu} + D_i$. Define the change:

$$\Delta\mathcal{F}_i = \mathcal{F}(\boldsymbol{\mu} + D_i,\, \mathbf{r} + e_i) - \mathcal{F}(\boldsymbol{\mu},\, \mathbf{r})$$

A spike reduces free energy if and only if $\Delta\mathcal{F}_i < 0$.

4.2 Expanding each term

Sensory term. Let $\boldsymbol{\epsilon}_y = y - C\boldsymbol{\mu}$. After the spike, $\boldsymbol{\epsilon}_y \to \boldsymbol{\epsilon}_y - CD_i$:

$$\Delta\mathcal{F}_i^{(y)} = -2\boldsymbol{\epsilon}_y^\top \Pi_y C D_i + D_i^\top C^\top \Pi_y C\, D_i$$

Prior term. Let $\boldsymbol{\epsilon}_\mu = \boldsymbol{\mu}^- - \boldsymbol{\mu}$. After the spike, $\boldsymbol{\epsilon}_\mu \to \boldsymbol{\epsilon}_\mu - D_i$:

$$\Delta\mathcal{F}_i^{(\mu)} = -2\boldsymbol{\epsilon}_\mu^\top \Pi_\mu D_i + D_i^\top \Pi_\mu D_i$$

Spike-cost term.

$$\Delta\mathcal{F}_i^{(\beta)} = \beta\,(2r_i + 1)$$

4.3 Collecting all contributions

Summing the three terms:

$$\Delta\mathcal{F}_i = -2D_i^\top\big[C^\top \Pi_y(y - C\boldsymbol{\mu}) - \Pi_\mu(\boldsymbol{\mu} - \boldsymbol{\mu}^-)\big] + D_i^\top M D_i + \beta(2r_i + 1)$$

The bracket simplifies:

$$C^\top \Pi_y(y - C\boldsymbol{\mu}) - \Pi_\mu(\boldsymbol{\mu} - \boldsymbol{\mu}^-) = C^\top \Pi_y y + \Pi_\mu \boldsymbol{\mu}^- - (C^\top \Pi_y C + \Pi_\mu)\boldsymbol{\mu} = \mathbf{q}_t - M\boldsymbol{\mu} = \mathbf{q}_t - MD\mathbf{r}$$

Hence:

$$\Delta\mathcal{F}_i = -2D_i^\top(\mathbf{q}_t - MD\mathbf{r}) + D_i^\top M D_i + \beta(2r_i + 1)$$

4.4 The spike rule

Requiring $\Delta\mathcal{F}_i < 0$ and dividing by 2 defines a natural membrane voltage and threshold:

Spike rule — neuron i fires when $v_i > T_i$

$$v_i = D_i^\top(\mathbf{q}_t - MD\mathbf{r}) = D_i^\top(\mathbf{q}_t - M\boldsymbol{\mu})$$
$$T_i = \tfrac{1}{2}\, D_i^\top M D_i + \beta\Big(r_i + \tfrac{1}{2}\Big)$$

The threshold has two components: a fixed base component $T_i^{\text{base}} = \frac{1}{2}O_{ii}$ where $O = D^\top M D$, and an activity-dependent component $\beta(r_i + \frac{1}{2})$ that implements automatic gain control.

Vector Voltage Form and Weight Matrices

Writing the voltages of all neurons simultaneously:

$$\mathbf{v} = D^\top \mathbf{q}_t - D^\top M D \mathbf{r} = D^\top C^\top \Pi_y y + D^\top \Pi_\mu \boldsymbol{\mu}^- - O\mathbf{r}$$

Voltage decomposition

$$\mathbf{v} = W_y\, y + W_p\, \boldsymbol{\mu}^- - O\,\mathbf{r}$$

$$W_y = D^\top C^\top \Pi_y \quad \text{(sensory feedforward weights)}$$

$$W_p = D^\top \Pi_\mu \quad \text{(prior input weights)}$$

$$O = D^\top M D = D^\top (C^\top \Pi_y C + \Pi_\mu) D \quad \text{(recurrent inhibitory weights)}$$

In the matched-channel case $C = I$ with channel-diagonal precisions, O simplifies to:

$$O = D^\top \operatorname{diag}(\pi_y + \pi_\mu)\, D$$

which is symmetric positive semidefinite. This symmetry is crucial: it guarantees that the spiking dynamics are gradient descent on a proper Lyapunov function.

Voltage Dynamics

6.1 Deriving the membrane ODE

Differentiating the voltage definition $\mathbf{v} = \mathbf{b} - O\mathbf{r}$ with respect to time, where $\mathbf{b} = W_y y + W_p \boldsymbol{\mu}^-$ is the bias current (held frozen within one time-step so $\dot{\mathbf{b}} = 0$), and using $\dot{\mathbf{r}} = -\lambda \mathbf{r} + \mathbf{s}$:

$$\dot{\mathbf{v}} = -O\dot{\mathbf{r}} = -O(-\lambda \mathbf{r} + \mathbf{s}) = \lambda O\mathbf{r} - O\mathbf{s}$$

Substituting $O\mathbf{r} = \mathbf{b} - \mathbf{v}$:

$$\dot{\mathbf{v}} = \lambda(\mathbf{b} - \mathbf{v}) - O\mathbf{s} = -\lambda \mathbf{v} + \lambda \mathbf{b} - O\mathbf{s}$$

Voltage dynamics (frozen ZOH window)

$$\dot{\mathbf{v}} = -\lambda \mathbf{v} + \lambda(W_y y + W_p \boldsymbol{\mu}^-) - O\mathbf{s}$$

The leak term $-\tau \mathbf{v}$ continuously drives the voltage toward the bias current; each spike $s_i = 1$ instantly inhibits the whole population by the column Oe_i.

6.2 Between-step voltage jump

At the transition between two outer time-steps, the bias $\mathbf{b}$ jumps because both y and $\boldsymbol{\mu}^-$ are refreshed. To maintain continuity of the algebraic free-energy interpretation without resetting $\mathbf{v}$ from scratch:

$$\mathbf{v}(t^+) \leftarrow \mathbf{v}(t^-) + (\mathbf{b}_{\text{new}} - \mathbf{b}_{\text{old}})$$

If additionally the precision matrices changed (e.g. after a slow learning update), the curvature matrix O also changes and an additional correction is needed:

$$\mathbf{v}(t^+) \leftarrow \mathbf{v}(t^-) + (\mathbf{b}_{\text{new}} - \mathbf{b}_{\text{old}}) - (O_{\text{new}} - O_{\text{old}})\mathbf{r}$$

This ensures the voltage remains the exact algebraic free-energy gradient even across discontinuous jumps in the inputs or precisions.

6.3 Stability

A potential source of confusion: writing the dynamics in terms of $\mathbf{r}$ gives an apparent $+\lambda O\mathbf{r}$ term that looks destabilising because $O \succ 0$. Rewriting in terms of $\mathbf{v} = \mathbf{b} - O\mathbf{r}$ reveals the standard negative-leak form $-\lambda\mathbf{v}$, which is globally stable. The positive definiteness of O is a coordinate artifact, not a stability hazard.

Complete Fast-Loop Summary

Full equations — SFEC fast loop (matched-channel, $C = I$)

Prior: $\boldsymbol{\mu}_t^- = \Phi\,\boldsymbol{\mu}_{t-1}, \quad \Phi \approx I + \Delta t\, A_{\text{int}}$

Information vector: $\mathbf{q}_t = \Pi_y y_t + \Pi_\mu\, \boldsymbol{\mu}_t^-$

Curvature: $M = \Pi_y + \Pi_\mu, \quad O = D^\top M D$

Voltage: $v_i = D_i^\top(\mathbf{q}_t - M\boldsymbol{\mu})$

Threshold: $T_i = \frac{1}{2}O_{ii} + \beta(r_i + \frac{1}{2})$

Dynamics: $\dot{\mathbf{v}} = -\lambda\mathbf{v} + \lambda\mathbf{b} - O\mathbf{s}, \quad \mathbf{b} = D^\top(\Pi_y y + \Pi_\mu\boldsymbol{\mu}^-) \in \mathbb{R}^N$

Spike counts: $\dot{\mathbf{r}} = -\lambda\mathbf{r} + \mathbf{s}, \quad \boldsymbol{\mu} = D\mathbf{r}$

Exact Bayes optimum ($\beta = 0$): $\boldsymbol{\mu}^\star = M^{-1}\mathbf{q}_t = \dfrac{\pi_y y + \pi_\mu\boldsymbol{\mu}^-}{\pi_y + \pi_\mu}$

> ***Dimensional note on the bias current.*** *The bias current* $\mathbf{b}$ *lives in* neuron *space* $\mathbb{R}^N$*, not in state space* $\mathbb{R}^n$*. Concretely,* $\mathbf{b} = D^\top(\Pi_y y + \Pi_\mu\boldsymbol{\mu}^-)$ *for* $C = I$*, and* $\mathbf{b} = D^\top(C^\top\Pi_y y + \Pi_\mu\boldsymbol{\mu}^-)$ *in the general* $C \neq I$ *case. The earlier voltage decomposition* $\mathbf{v} = W_y y + W_p\boldsymbol{\mu}^- - O\mathbf{r}$ *already carries the* $D^\top$ *correctly inside* $W_y = D^\top C^\top\Pi_y$ *and* $W_p = D^\top\Pi_\mu$*; the boxed summary makes this explicit.*

The exact Bayesian posterior is thus a simple channel-wise weighted average. The spiking network approximates this optimum through local threshold crossings. When the precisions are correct, experiments confirm that the spiking estimate $\boldsymbol{\mu}$ tracks the exact Bayesian posterior $\boldsymbol{\mu}^\star$ very closely: this is the sense in which *the fast loop works*.

8.1 What needs to be learned

For each channel k, the optimal Kalman gain is

$$K_k^\star = \frac{P_k^-}{P_k^- + R_k}$$

where P_k^- is the prior (predictive) variance and R_k is the sensor noise variance. Rather than estimating P^- and R separately, the slow loop learns:

1. The *gain* $K_k = \sigma(g_k) \in (0, 1)$, where σ is the sigmoid function and g_k is an unconstrained logit parameter.
2. The *total innovation scale* $S_k = P_k^- + R_k$, the total variance of the innovation stream.

From these two scalars the precisions are reconstructed as:

$$q_{\mu,k} = K_k S_k \approx P_k^-, \qquad q_{y,k} = (1 - K_k) S_k \approx R_k, \qquad \pi_{y,k} = q_{y,k}^{-1}, \quad \pi_{\mu,k} = q_{\mu,k}^{-1}$$

8.2 The innovation

For each channel k, the *innovation* ν_k is the pre-fit prediction error:

$$\nu_{t,k} = y_{t,k} - \mu_{t,k}^-$$

Crucially, the innovation is measured *before* the posterior correction; it is therefore less contaminated by the current precision settings than the post-fit residuals.

8.3 Why optimal filters produce white innovations

This is the central statistical fact underlying the slow learning rule.

The true state dynamics and measurement model are:

$$x_t = \phi\, x_{t-1} + w_{t-1}, \qquad y_t = x_t + \eta_t$$

with $w \sim \mathcal{N}(0, Q)$, $\eta \sim \mathcal{N}(0, R)$, all mutually independent. (The scalar notation suppresses the channel index k.)

The filter's prior prediction is $\mu^- = \phi\, \mu_{t-1}$ and after observing y_t:

$$\mu_t = \mu^- + K\nu_t = \mu^- + K(y_t - \mu^-)$$

> ***Scope of the scalar derivation.*** *The scalar recursion above, and the per-channel whitening identity derived from it, are exact only when the channels are dynamically decoupled — that is, when* $\Phi = \mathrm{diag}(\phi_1, \ldots, \phi_n)$ *and both* Q *and* R *are diagonal. If* A_{int} *has off-diagonal entries, then* Φ *couples the channels, and each channel's innovation depends on other channels' prediction errors. In that regime the per-channel rule* $c_{1,k} \propto K_k^\star - K_k$ *is an approximation. The exact whitening condition becomes matrix-valued:* $\Gamma_1 = \mathbb{E}[\nu_t \nu_{t-1}^\top] = \Phi(K^\star - K)S$*, where* K *is now a gain* matrix*. A coupled-channel generalisation of the slow loop — replacing the scalar traces with* $n \times n$ *matrix exponential moving averages and updating a full gain matrix — is a natural direction for future implementations.*

8.4 The error recursion

Lemma 1 (Error recursion)**.** *Let $e_t^- = x_t - \mu_t^-$ be the prior prediction error. Then:*

$$e_t^- = \phi(1-K)e_{t-1}^- + w_{t-1} - \phi K \eta_{t-1}$$

Proof. **Part 1: Prior error in terms of yesterday's final error.**
By definition:

$$e_t^- = x_t - \mu_t^- = (\phi x_{t-1} + w_{t-1}) - \phi\mu_{t-1} = \phi(x_{t-1} - \mu_{t-1}) + w_{t-1} = \phi\, e_{t-1} + w_{t-1}$$

Part 2: Yesterday's final error e_{t-1}.
The posterior update was:

$$\mu_{t-1} = \mu_{t-1}^- + K\nu_{t-1} = \mu_{t-1}^- + K(y_{t-1} - \mu_{t-1}^-) = \mu_{t-1}^- + K(e_{t-1}^- + \eta_{t-1})$$

Hence:

$$e_{t-1} = x_{t-1} - \mu_{t-1} = x_{t-1} - \mu_{t-1}^- - K(e_{t-1}^- + \eta_{t-1}) = e_{t-1}^- - Ke_{t-1}^- - K\eta_{t-1} = (1-K)e_{t-1}^- - K\eta_{t-1}$$

Part 3: Assembly.
Substituting into the Part 1 result:

$$e_t^- = \phi\big[(1-K)e_{t-1}^- - K\eta_{t-1}\big] + w_{t-1} = \phi(1-K)e_{t-1}^- + w_{t-1} - \phi K\eta_{t-1}$$

□

Reading the error recursion aloud: today's prediction error has three contributions:

- $\phi(1-K)e_{t-1}^-$: the ghost of yesterday's prediction error, attenuated by the gain. Larger K wipes out more of the inherited error.
- $+w_{t-1}$: the new, unpredictable process noise that arrived overnight.
- $-\phi K\eta_{t-1}$: the sensor-noise *hangover*: because we trusted the sensor by an amount K, we absorbed sensor noise η_{t-1} into our belief and it now propagates forward.

8.5 Lag-one innovation covariance

The innovation at time t is $\nu_t = e_t^- + \eta_t$. Substituting the error recursion:

$$\nu_t = \big[\phi(1-K)e_{t-1}^- + w_{t-1} - \phi K\eta_{t-1}\big] + \eta_t$$

Yesterday's innovation was $\nu_{t-1} = e_{t-1}^- + \eta_{t-1}$.

Proposition 1 (Lag-one covariance)**.** *Under the scalar matched-channel model with fixed gain K running at stationarity, let P_K^- denote the stationary prior-error variance induced by that gain. Then:*

$$\mathbb{E}[\nu_t\nu_{t-1}] = \phi\, S_K\left(K_K^\star - K\right)$$

where $S_K = P_K^- + R$ is the total innovation variance under gain K, and $K_K^\star = P_K^- / S_K$ is the gain that would be optimal given the current P_K^-.

Proof. Expand $\mathbb{E}[\nu_t\,\nu_{t-1}] = \mathbb{E}\big[(\phi(1-K)e^-_{t-1} + w_{t-1} - \phi K\eta_{t-1} + \eta_t)(e^-_{t-1} + \eta_{t-1})\big]$.

Because w, η, and e^- are mutually independent and zero-mean, all cross-terms between *different* random variables vanish. Specifically:

$$\begin{aligned} \mathbb{E}[w_{t-1}\,e^-_{t-1}] &= 0 & \qquad \mathbb{E}[w_{t-1}\,\eta_{t-1}] &= 0 \\ \mathbb{E}[\eta_t\,e^-_{t-1}] &= 0 & \qquad \mathbb{E}[\eta_t\,\eta_{t-1}] &= 0 \\ \mathbb{E}[\eta_{t-1}\,e^-_{t-1}] &= 0 \end{aligned}$$

Only two terms survive — those where a variable multiplies itself:

$$\mathbb{E}[\nu_t\nu_{t-1}] = \phi(1-K)\,\mathbb{E}[(e^-_{t-1})^2] - \phi K\,\mathbb{E}[\eta^2_{t-1}] = \phi(1-K)P^-_K - \phi KR$$

Expanding:

$$= \phi P^-_K - \phi K P^-_K - \phi KR = \phi P^-_K - \phi K(P^-_K + R) = \phi S_K\left(\frac{P^-_K}{S_K} - K\right) = \phi S_K\,(K^\star_K - K)$$

□

> *This equation is the entire heart of the learning algorithm. It says that the lag-one autocorrelation of the innovation stream is a signed, scaled measure of how far the current gain K is from the optimal gain $K^\star$:*
>
> - *If $K < K^\star$: the filter under-trusts the sensor; innovations are* positively correlated*; the gain should be increased.*
> - *If $K > K^\star$: the filter over-trusts the sensor; innovations are* negatively correlated*; the gain should be decreased.*
> - *If $K = K^\star$: innovations are uncorrelated (*white*); the filter is optimal.*
>
> *The zero crossing is the unique fixed point where the innovation sequence is white.*

8.6 From discrete statistics to continuous-time traces

A real-time neural network cannot compute ensemble expectations. Instead, it approximates them using *leaky integrators*, which are exponential moving averages implemented in continuous time.

For a differential equation of the form $\tau\dot{c} = -c + x$, the steady-state solution is the exponential moving average:

$$c(t) = \int_0^t \frac{1}{\tau} e^{-(t-s)/\tau} x(s)\,\mathrm{d}s$$

The weighting kernel $\frac{1}{\tau}e^{-(t-s)/\tau}$ integrates to 1, so this is a proper moving average. The time constant τ sets the integration window.

8.7 The three local traces

For each channel k, the slow loop maintains three local state variables.

8.7.1 *Delay trace z_k*

$$\tau_d \, \dot{z}_k = -z_k + \nu_k$$

The purpose of z_k is to hold a copy of the *previous* innovation $\nu_{t-1,k}$ so that the product $\nu_{t,k} \, z_k$ estimates the lag-one correlation.

8.7.2 *Innovation power trace $c_{0,k}$*

$$\tau_0 \, \dot{c}_{0,k} = -c_{0,k} + \nu_k^2$$

The leaky integrator accumulates the squared innovation. At steady state:

$$c_{0,k} \approx \mathbb{E}[\nu_k^2] = S_k = P_k^- + R_k$$

This is the continuous-time estimate of the total innovation variance, which is used directly as the learned scale.

8.7.3 *Lag-one covariance trace $c_{1,k}$*

$$\tau_1 \, \dot{c}_{1,k} = -c_{1,k} + \nu_k \, z_k$$

The leaky integrator accumulates the product of the current innovation ν_k and the delayed trace z_k. Provided the update ordering described above is observed, z_k holds $\nu_{t-1,k}$ when the product is formed, so at steady state:

$$c_{1,k} \approx \mathbb{E}[\nu_{t,k} \, \nu_{t-1,k}] = \phi_k S_k (K_k^\star - K_k)$$

So $c_{1,k}$ tracks the lag-one covariance and its sign reveals the gain error (see Proposition 1 and the surrounding note on P_K^- dependence).

8.7.4 *Summary*

Local innovation traces (per channel k) — update order matters

$$\tau_0 \, \dot{c}_{0,k} = -c_{0,k} + \nu_k^2 \qquad c_{0,k} \approx S_k \qquad \text{(innovation power} \approx \text{total variance)}$$

$$\tau_1 \, \dot{c}_{1,k} = -c_{1,k} + \nu_k \, z_k \quad \text{where } z_k \text{ holds } \nu_{t-1,k} \quad c_{1,k} \approx \phi_k S_k (K_k^\star - K_k)$$

$$z_k \leftarrow \nu_k \qquad \text{(delay register — written } \textit{after} \text{ forming the lag product)}$$

9.1 Gain update

The gain is parameterised as $K_k = \sigma(g_k)$ where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid function. Learning in logit space automatically enforces $K_k \in (0, 1)$.

The gradient signal from Proposition 1 is:

$$c_{1,k} \propto K_k^\star - K_k$$

A positive $c_{1,k}$ means $K_k < K_k^\star$: increase the gain. In the scalar matched-channel case, the system dynamics are captured by $\phi_k = \Phi_{kk}$ (the diagonal of the discretised state-transition matrix). The sign of ϕ_k is positive for all stable physical systems, so it acts only as a scaling.

The normalised gain drive for channel k is therefore:

$$\text{drive}_k = \text{sgn}(\phi_k)\, \frac{c_{1,k}}{\max(c_{0,k},\, \varepsilon)}$$

where $\varepsilon > 0$ is a small floor preventing division by near-zero. The normalisation by $c_{0,k} \approx S_k$ removes the dependence on the total innovation amplitude, leaving a signal that measures *only* the fractional gain error.

The logit update is then:

$$\dot{g}_k = \eta_\lambda\, \text{drive}_k - \kappa_\mu (g_k - g_k^0)$$

The first term drives g_k (and hence K_k) toward $K_k^\star$. The second term is a weak regularisation toward an initial guess g_k^0, preventing divergence if the innovation trace is noisy.

9.2 Scale update

The total innovation scale estimate is updated toward $c_{0,k}$:

$$\dot{S}_k = \frac{1}{\tau_\lambda}(c_{0,k} - S_k) - \kappa_y (S_k - S_k^0)$$

where τ_λ is the slow learning time constant, κ_y is a weak regularisation weight, and S_k^0 is an initial scale guess. The update is clipped in log-space to prevent violent scale jumps:

$$\Delta \log S_k = \text{clip}(\log(S_k^{\text{new}}) - \log(S_k),\ -\delta_{\max},\ +\delta_{\max})$$

9.3 Reconstructing the precisions

Once K_k and S_k are updated, the channel precisions are reconstructed as:

$$q_{\mu,k} = K_k S_k \approx P_k^-, \qquad q_{y,k} = (1 - K_k) S_k \approx R_k$$
$$\pi_{\mu,k} = \frac{1}{q_{\mu,k}}, \qquad \pi_{y,k} = \frac{1}{q_{y,k}}$$

These are then injected into the fast loop to update O, the bias weights W_y, W_p, and the thresholds T_i^{base}.

9.4 Settled-gate condition

Trace updates are multiplied by a binary gate that equals 1 only when the spiking population has settled (i.e. the voltage is below all thresholds and the free energy has not increased since the start of the inference step). This prevents the slow learner from accumulating false statistics during transient, unsettled inference:

$$\text{gate} = \mathbf{1}\left[\max_i (v_i - T_i) \le 0 \;\text{ and }\; \mathcal{F}_{\text{end}} \le \mathcal{F}_{\text{start}}\right]$$

9.5 Full slow-loop summary

Innovation-whitening slow loop (per channel k)

Innovation: $\nu_k = y_k - \mu_k^-$

Delay trace: $\tau_d \dot{z}_k = -z_k + \nu_k$

Power trace: $\tau_0 \dot{c}_{0,k} = -c_{0,k} + \nu_k^2$

Lag-one trace: $\tau_1 \dot{c}_{1,k} = -c_{1,k} + \nu_k z_k$

Gain drive: $\text{drive}_k = \text{sgn}(\phi_k)\, c_{1,k} / \max(c_{0,k}, \varepsilon)$

Logit update: $\dot{g}_k = \eta_\lambda \,\text{drive}_k - \kappa_\mu (g_k - g_k^0)$

Gain: $K_k = \sigma(g_k) = 1/(1 + e^{-g_k})$

Scale update: $\dot{S}_k = \frac{1}{\tau_\lambda}(c_{0,k} - S_k) - \kappa_y (S_k - S_k^0)$

Reconstruct: $q_{\mu,k} = K_k S_k, \quad q_{y,k} = (1 - K_k) S_k, \quad \pi_{\mu,k} = 1/q_{\mu,k}, \quad \pi_{y,k} = 1/q_{y,k}$

Assumptions and Their Possible Relaxation

The derivation above rests on several simplifying assumptions. We discuss each in turn.

10.1 Matched-channel observation ($C = I$, $m = n$)

What it buys. With $C = I$ the free-energy errors $e_y = y - \mu$ and $e_\mu = \mu^- - \mu$ both live in the same n-dimensional channel space. Precisions can be diagonal and channel-decomposable; the curvature matrix simplifies to $O = D^\top \text{diag}(\pi_y + \pi_\mu) D$; and the lag-one covariance identity holds separately for each scalar channel.

How to relax it. For a general observation matrix $C \in \mathbb{R}^{m \times n}$ with $m < n$, the sensory error $e_y = y - C\mu$ lives in measurement space while the prior error $e_\mu = \mu^- - \mu$ lives in state space. The information fusion becomes:

$$\boldsymbol{\mu}^\star = (C^\top \Pi_y C + \Pi_\mu)^{-1} (C^\top \Pi_y y + \Pi_\mu \boldsymbol{\mu}^-)$$

The fast loop generalises immediately by replacing $W_y = D^\top C^\top \Pi_y$ and $O = D^\top (C^\top \Pi_y C + \Pi_\mu) D$.

The slow loop is harder to generalise. The innovation is still $\nu = y - C\mu^-$, which now lives in measurement space $\mathbb{R}^m$. The whiteness condition $\mathbb{E}[\nu_t \nu_{t-1}^\top] = 0$ becomes a matrix equation. A generalised learning rule would need to whiten the m-dimensional innovation process, requiring a gain matrix $K \in \mathbb{R}^{n \times m}$ instead of a

scalar. This requires learning nm parameters rather than n, complicating the local trace structure.

10.2 Diagonal precision matrices

What it buys. Diagonal Π_y and Π_μ make the free energy channel-separable: each channel k contributes independently, and the gain is a scalar per channel. The lag-one covariance identity holds channel-by-channel.

How to relax it. Full covariance matrices Π_y and Π_μ would allow the estimator to capture inter-channel correlations in the noise. The free-energy functional and spike condition remain valid for any positive-definite Π_y and Π_μ. The voltage equation becomes $\mathbf{v} = D^\top C^\top \Pi_y y + D^\top \Pi_\mu \boldsymbol{\mu}^- - O\mathbf{r}$ with $O = D^\top (C^\top \Pi_y C + \Pi_\mu) D$, which is unchanged in form.

The slow loop, however, now needs to learn a full matrix rather than scalars. In principle the whiteness condition generalises: $K_{\text{full}} = P^-(P^- + R)^{-1}$ is the matrix Kalman gain and the whiteness condition is $\mathbb{E}[\nu_t \nu_{t-1}^\top] = 0$ for the matrix-valued innovation. Implementing this with local synaptic traces is an open research problem.

10.3 Linear, time-invariant dynamics

What it buys. The error recursion (Lemma 1) and the lag-one covariance identity (Proposition 1) rely on ϕ being a scalar constant. LTI dynamics make the noise statistics stationary and the optimal gain constant over time.

How to relax it. For a time-varying or nonlinear system, ϕ_t varies across time steps and the optimal gain becomes a time-varying quantity $K_t^\star$. The innovation-whitening condition still applies instantaneously, but the traces c_0 and c_1 must track a non-stationary distribution. Adaptive time constants (e.g. shortening τ_0 and τ_1 when the innovation variance changes rapidly) would be needed.

10.4 Slow-timescale separation

What it buys. The derivation assumes that the precision parameters K_k and S_k change on a time scale $\tau_\lambda \gg \tau_0 \gg \tau$ (slow learner, medium trace integration, fast inference). Under this separation, c_0 and c_1 are approximately converged when the slow update fires, making the learning rule equivalent to a gradient step on a stationary objective.

How to relax it. If τ_λ is not much larger than τ_0, the traces are still adapting when the gains are updated: the learner is chasing a moving target. In that regime, the rule becomes an online stochastic gradient algorithm (analogous to least-mean-squares) and convergence is guaranteed only in a mean-squared sense with a small enough learning rate. The regularisation terms $\kappa_y(S - S^0)$ and $\kappa_\mu(g - g^0)$ help stabilise this regime.

10.5 Scalar channel decomposition

What it buys. The whole SFEC construction assumes that the decoder matrix D maps faithfully from the overcomplete spike space to state space and back. With

$C = I$ and diagonal precisions, each of the n channels is perfectly decoupled in the free-energy functional. This makes the problem separable and allows one independent slow learner per channel.

How to relax it. If off-diagonal precision entries exist, or if C mixes channels, then a single channel's innovation carries information about multiple state components. In that case, learning rules would need to operate on multivariate statistics, coupling the slow learners across channels.

What Was Tried and Why It Failed

The innovation-whitening approach was arrived at after a sequence of well-motivated but ultimately unsuccessful attempts. Each failure taught a concrete lesson. We document them here to serve as a map of the design space.

11.1 Neuron-space local precision learner

Approach

Allow each coding neuron i to learn its own local gain or precision, applied as a row-wise scaling in neuron space: $O_{\text{eff}} = \Pi_N O$ with Π_N diagonal in neuron index space.

What went wrong. The original recurrent matrix $O = D^\top MD$ is symmetric positive semidefinite whenever M is. Applying a diagonal neuron-space gain Π_N from the left destroys symmetry unless all gains are equal. An asymmetric O_{eff} means the voltage dynamics are no longer gradient descent on any well-defined energy surface. The network begins to rotate and thrash rather than relax; spike rates diverge; the slow learner observes enormous artificial errors and prior precision collapses.

Key lesson. Precision must be placed in *channel space*, not in neuron space, to preserve the symmetric positive semidefinite structure of the curvature matrix.

11.2 Channel-space two-precision learner from post-fit residuals

Approach

Learn q_y from the observation residual $e_y = y - \mu$ and q_μ from the prior residual $e_\mu = \mu^- - \mu$, both measured after the posterior correction.

What went wrong. The posterior residuals are algebraically determined by the current precision ratio:

$$e_y = \frac{\pi_\mu}{\pi_y + \pi_\mu}\nu, \qquad e_\mu = \frac{\pi_y}{\pi_y + \pi_\mu}\nu$$

Any learner that reads both sides from the same-step posterior is reading its own output. It converges to a *self-consistent* fixed point — one where the residuals happen to match the assumed variances — rather than to the Bayesian-optimal fixed point. Empirically: the learned quantities agree with each other but not with the true injected noise; sweeping the true R does not change the learned $\hat{R}$.

Key lesson. If both precision parameters are learned from same-step posterior residuals, the learner identifies itself rather than the true noise model.

11.3 Variational-Bayes / active-inference whitening on post-fit residuals

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Approach

Apply the standard VB update for Gaussian precision: target $\hat{\pi}_y = 1/\langle e_y^2 \rangle$ and $\hat{\pi}_\mu = 1/\langle e_\mu^2 \rangle$. This is the correct M-step in a static Gaussian model.

What went wrong. In the static Gaussian model, the E-step and M-step are decoupled because the sufficient statistics are exact. In our estimator, the same-step posterior changes whenever the precision changes, so the M-step update is applied to statistics that were generated under the *previous* precision — but not the previous external reality. The effect is the same as the above: the learner homes in on self-consistency rather than truth. Additionally, post-fit residuals are compressed versions of the innovation; they carry less information about the noise split.

Key lesson. VB whitening of post-fit residuals is too endogenous for an online sequential estimator where the posterior continuously depends on the precision being learned.

11.4 Ratio-only / log-ratio learner

Approach

Note that the common scale $S = P^- + R$ partially cancels in the spike condition (since both precisions scale together). Learn only the ratio $\rho = \pi_y / \pi_\mu$ (or equivalently the logit of the gain g) and hold the total scale fixed.

What went wrong (two reasons). *First*, the statistic driving the ratio update was still derived from the current posterior, so the same endogenous-loop problem applied. The learner recovered the filter's own ratio rather than the Bayesian-optimal ratio.

Second, even if the ratio were identifiable, discarding the absolute scale causes long-run problems. The scale S_k determines the actual uncertainty magnitude reported to downstream systems and calibrates the trade-off between the regularisation prior g^0 and the data. A ratio-only learner became "self-consistent but not physically identified."

Key lesson. Both the gain (ratio) and the scale (total variance) must be learned, and they require distinct signal sources: pre-fit innovations for the scale and lag-one correlations for the gain.

11.5 Blockwise local-variance learner

Approach

Freeze precision parameters over a window of T steps. Within the window, accumulate sufficient statistics and then do a closed-form M-step update. This removes the moving-target problem because the posterior family is fixed during accumulation.

What improved. Freezing the window genuinely helped: the learner was no longer chasing a moving posterior. Observation-noise estimation improved substantially over the purely online rules.

What still failed. The prior-side estimate $\hat{P}^- \approx \langle \delta^2 \rangle + \hat{P}_{\text{post}}$ (where $\delta = \mu - \mu^-$ is the posterior correction) was fragile under the SNN's finite relaxation, batched spike dynamics, and diagonal approximation. In high-activity regimes the estimate was dominated by the SNN's own firing statistics rather than the physical process noise. Furthermore, the observation-side estimate remained contaminated by the current precision split.

Key lesson. Blockwise learning helps considerably by removing the moving-target problem, but the choice of sufficient statistics still dominates performance. Post-fit statistics on the prior side are too noisy for reliable prior-variance estimation.

11.6 Observation-side innovation fix

Approach

Replace the post-fit observation residual with the pre-fit innovation $\nu = y - \mu^-$ to estimate R, using $\mathbb{E}[\nu^2] = P^- + R$ and a separate estimate of P^- to isolate R.

What improved. Because ν is measured before the posterior correction, it is less entangled with the current precision split. The observation-noise identification improved markedly, particularly on sweeps of R.

What still failed. The estimate of R still depended on a separate estimate of P^-. If the prior-side learner was inaccurate, the observation estimate inherited that error. Jointly sweeping both Q and R revealed that the two estimates remained coupled.

Key lesson. Pre-fit innovations are the correct signal for the *total* scale $S = P^- + R$. Separating P^- and R from innovations alone is not possible without additional assumptions; hence the learning rule uses S as a single object and infers the gain K (which captures the ratio P^-/S) separately via the whiteness condition.

11.7 Probe-based gain identifier

Approach

Inject a small known probe signal p_t into the sensory channel and identify the current Kalman gain from the ratio of probe cross-correlation to probe power:

$$\hat{K} = \frac{\mathbb{E}[\delta_t p_t]}{\mathbb{E}[p_t^2]}, \quad \hat{S} = \mathbb{E}[\tilde{\nu}^2] - \mathbb{E}[p^2], \quad \hat{P}^- = \hat{K}\hat{S}, \quad \hat{R} = (1 - \hat{K})\hat{S}$$

where $\tilde{\nu}$ is the innovation with the probe injected and $\delta = \mu - \mu^-$ is the posterior correction.

What went wrong. The probe cross-correlation identifies the *current filter's* gain, not the Bayesian-optimal gain. Specifically, since $\delta \approx K_{\text{current}}(\nu + p)$,

$$\frac{\mathbb{E}[\delta p]}{\mathbb{E}[p^2]} \approx K_{\text{current}}$$

Under correct precisions, this coincidentally returns the right answer. Under wrong precisions, it faithfully returns the wrong, currently-imposed gain. The probe rule therefore had no ability to self-correct: it was a *self-identification* rule for the current filter, not a *truth-identification* rule.

Key lesson. Any signal derived from the posterior update step is corrupted by the current filter parameters. Only signals derived before the update — like the innovation — can provide genuinely exogenous information about the underlying noise statistics.

Summary and Conclusions

12.1 What works and why

The innovation-whitening approach succeeds because it separates what the filter knows ($\boldsymbol{\mu}^-$, the prior) from what it observes (y, the measurement), and exploits the statistical property that the *difference* $\nu = y - \boldsymbol{\mu}^-$ should be serially uncorrelated under the optimal filter.

The three traces z, c_0, c_1 are each *local* in the sense that they require only quantities available at the channel level (ν_k), no communication across channels, and no memory beyond a simple leaky integrator. This is biologically plausible and numerically inexpensive.

12.2 Why previous approaches failed

A single underlying failure mode explains all the failed learners: *endogeneity*. Any learning signal derived from the posterior mean μ or the posterior residuals e_y, e_μ is algebraically coupled to the current precision parameters. The learner sees its own reflection rather than the external world.

The solution is to measure the world before the posterior correction — i.e. to use the innovation $\nu = y - \mu^-$ — and to exploit its *temporal structure* (lag-one covariance) rather than just its magnitude.